Parental occupational exposure to pesticides as risk factor for brain tumors in children and young adults: a systematic review and meta-analysis.

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OBJECTIVE: To examine the potential association between parental occupational exposure to pesticides and the occurrence of brain tumors in children and young adults. METHODS: Studies identified from a MEDLINE search through 15 January 2013 and from the reference lists of identified publications were submitted to a systematic review and meta-analysis. Relative risk estimates were extracted from 20 studies published between 1974 and 2010. Most of the retrieved studies involved farm/agricultural jobs. Summary ratio estimates (SR) were calculated according to fixed and random-effect meta-analysis models. Separate analyses were conducted after stratification for study design, exposure parameters, disease definition, geographic location and age at diagnosis. RESULTS: Statistically significant associations were observed for parents potentially exposed to pesticides in occupational settings and the occurrence of brain tumor in their offspring after combining all case-control studies (summary odds ratio [SOR]: 1.30; 95%: 1.11, 1.53) or all cohort studies (summary rate ratio [SRR]: 1.53; 95% CI: 1.20, 1.95). Significantly increased risks were seen for prenatal exposure windows, for either exposed parent, for exposure defined as to pesticides as well as by occupational/industry title, for astroglial brain tumors and after combining case-control studies from North America or cohort studies from Europe. CONCLUSIONS: This meta-analysis supports an association between parental occupational exposure to pesticides and brain tumors in children and young adults, and adds to the evidence leading to the recommendation of minimizing (parental) occupational exposure to pesticides. These results must, however, be interpreted with caution because the impact of work-related factors others than pesticide exposure is not known.